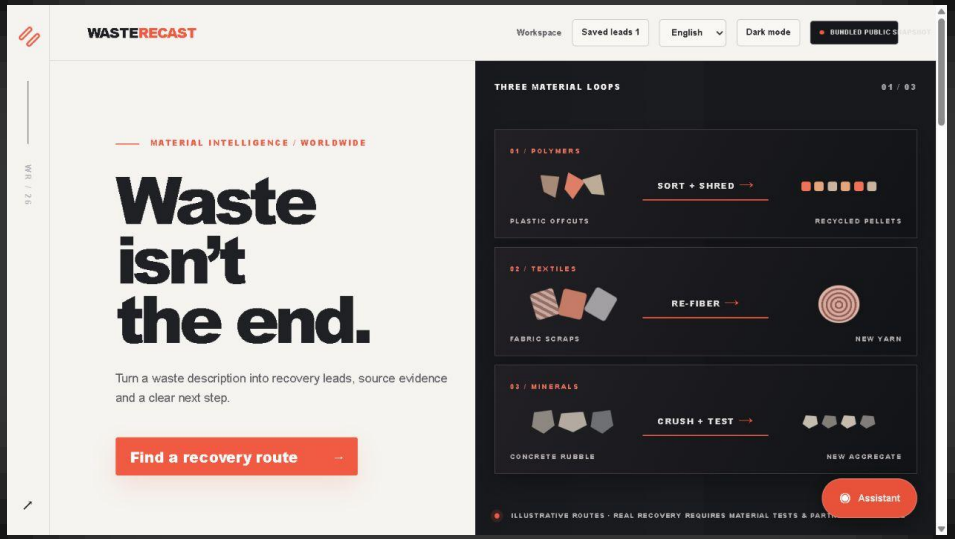
Google Cloud H2S

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Team Details
Team name: Clorex | Product: WasteRecast
Team leader: Zuhaib Ahmed
Members: Anshuman Pandey · Aliyan Ahmed
Problem statement: Sustainability & Social Impact
AI-assisted waste recovery and reuse discovery



WasteRecast

Turn a waste description into recovery leads, source evidence and a clear next step.

Built by Clorex

Brief about the idea

Waste is a potential input. Finding a usable route takes evidence.

THE BUSINESS PROBLEM

A generator knows its waste, but not which facility can accept its exact composition, quantity and condition.

OUR WORKING SOLUTION

Gemini extracts the material profile. WasteRecast screens public records, explains gaps and prepares an acceptance enquiry.

Theme: Sustainability & Social Impact — resource efficiency and informed decisions. No completed recovery or emissions savings claimed.

Opportunities

How is this different?

Evidence stays beside each lead. Missing specifications and unconfirmed acceptance stay visible.

How will it solve the problem?

Move from an unstructured description to a short, source-linked enquiry that a recycler can actually evaluate.

Our proposed USP

Recast combines conversational app control with a material-specific verification workflow—not a promise of a confirmed buyer.

Marketplaces and reuse exchanges already exist. Our differentiation is the workflow; uniqueness and commercial impact still need pilot validation.

List of features offered by the solution

01 / MATERIAL PROFILE

Text, optional PDF or image → structured Gemini profile, quoted evidence and missing facts. 16 material families.

02 / CLEAR RECOVERY LEADS

912 source-attributed India board records; worldwide open-map discovery. Five visible leads at a time.

03 / VERIFIABLE NEXT STEPS

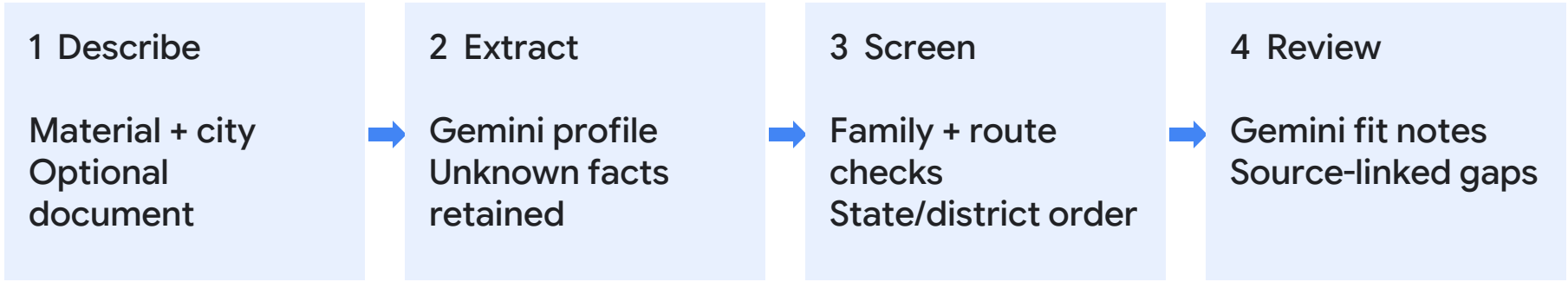
Source documents, published contacts where available, saved leads, four confirmation checks and enquiry export.

04 / ACCESSIBLE INTERACTION

Recast Live assistant with validated app tools; English, Hindi, Marathi and Spanish controls; light/dark themes.

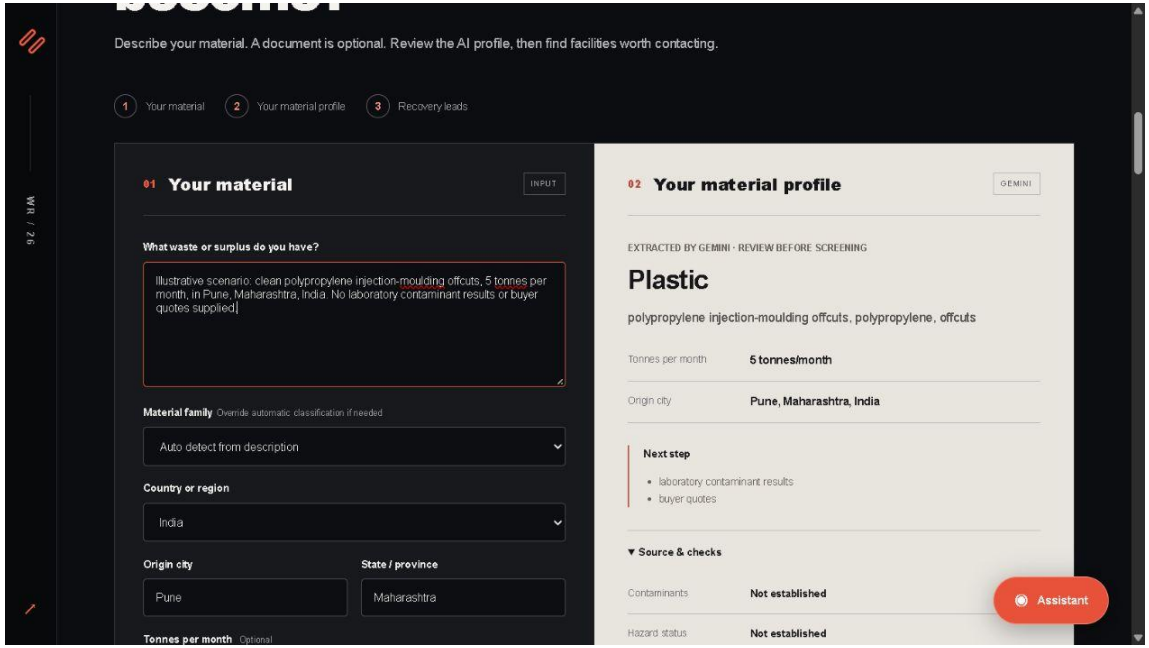
Public records are discovery evidence, not verified demand. Worldwide map coverage is best effort. Some diagnostics remain in English.

Process flow diagram or Use-case diagram



AI interprets and explains; deterministic checks and calculations remain inspectable. A human confirms acceptance before any movement.

Wireframes/Mock diagrams of the proposed solution (optional)

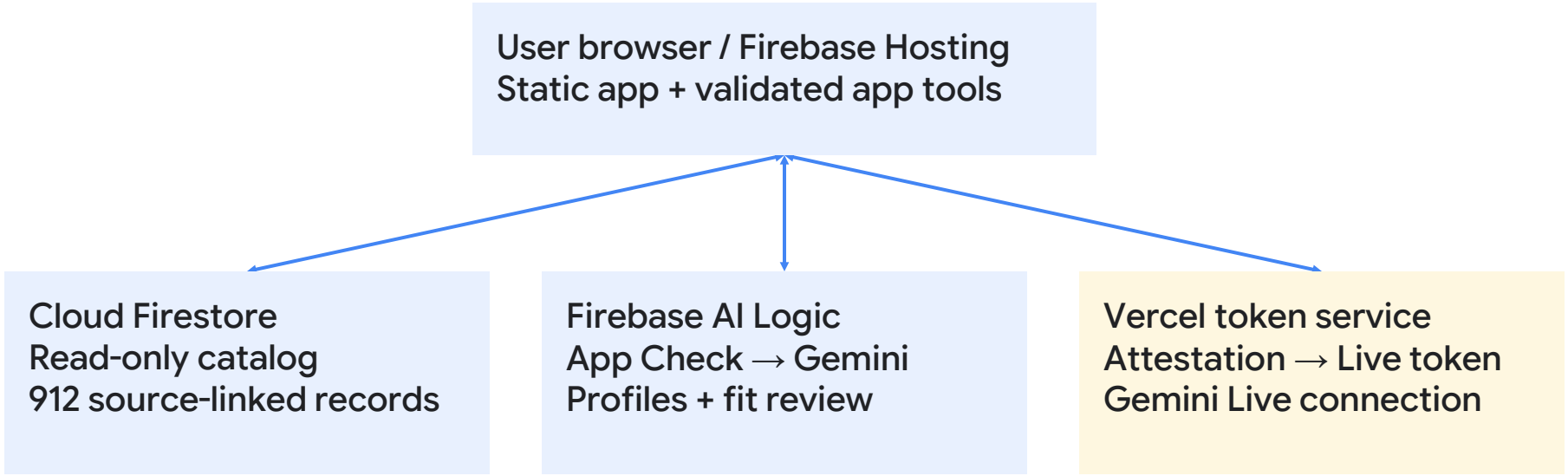


ACTUAL INTERFACE

1. Describe your material
2. Review the profile
3. Find recovery leads

Actual prototype, not a wireframe. AI walkthrough captured using the private local connection; public site uses Firebase AI Logic.

Architecture diagram of the proposed solution



Catalog imports: public board PDFs. Worldwide discovery: browser → open geocoding/maps.
Live audio: browser → Gemini directly after the token is issued.

Live deployment uses Firebase, not Cloud Run. Optional Cloud Run backend is a future path. Free quotas and third-party uptime apply.

Technologies to be used in the solution

Firebase Hosting	Public HTTPS web app on Spark. Static delivery with a sourced catalog fallback.
Cloud Firestore	912 public records in 23 read-only chunks; client writes denied.
Gemini + Firebase AI Logic	Structured extraction, evidence-aware fit review and conversational reasoning. App Check enforced.
Gemini Live + Vercel	Aoede speech and validated tool calls. Private key issues short-lived, one-use tokens.
Supporting stack	Vite / JavaScript; official board PDFs; OpenStreetMap / Overpass; Open-Meteo geocoding.

Cloud Run, Cloud SQL, BigQuery and Vertex AI are not deployed. Firebase is the permitted hosting option selected for this release.

Estimated implementation cost (optional)

Current prototype: configured for free tiers.

CURRENT SETUP

Firebase Spark + free Gemini quota + existing Vercel project. No Google Cloud billing account linked.

SCALING APPROACH

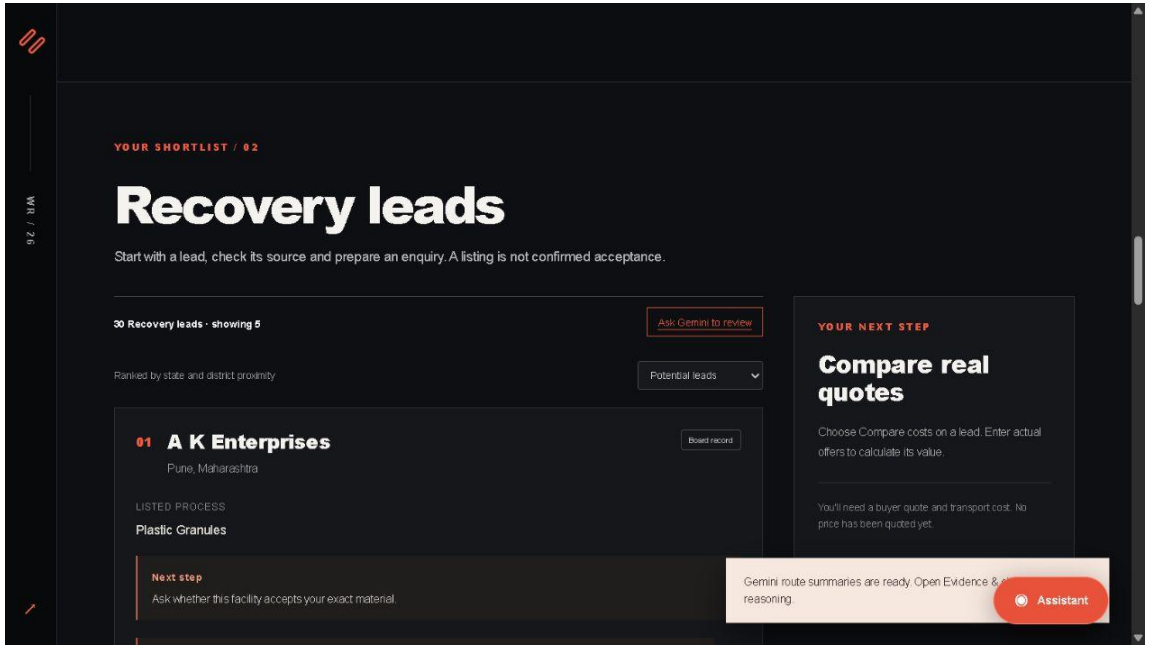
Static delivery; chunked reads; bounded AI calls, response cache and clear manual fallback when providers are unavailable.

Before a paid rollout

Measure requests, tokens, database reads and audio minutes. Add user authentication, shared quotas and monitoring before changing billing.

No cost or unlimited-availability guarantee. A paid production estimate requires measured traffic and current provider pricing.

Snapshots of the prototype



LEADS → ACTION

Real source records

A focused acceptance enquiry

No invented buyer prices

Illustrative polypropylene scenario + real public listings. Screenshots show the working local AI connection; hosted UI is the same application.

Prototype Performance report/Benchmarking

39 / 39 TESTS PASS

Automated regression tests cover screening, missing quotes, language keys, tool validation, microphone PCM and token controls.

GEMINI BACKEND / PASS

Live Firebase AI Logic request returned HTTP 200 and a structured polypropylene profile on 17 October 2026.

BROWSER WALKTHROUGH

Local AI profile, fit review and enquiry flow verified. Participant confirmed the public material profile works in Chrome/Edge.

NOT YET BENCHMARKED

No independent extraction-accuracy study, load test, verified buyer conversion or measured environmental savings.

Test counts describe automated checks, not AI accuracy. Codex's in-app browser could not fetch public AI; Chrome/Edge check was participant-reported.

Additional Details/Future Development (if any)

NEXT / BUYER ACCEPTANCE

Onboard consenting recyclers; collect actual accepted specifications, permits, capacity and written quotes.

PILOT / PROVE THE VALUE

Measure time to qualified enquiry, acceptance rate and verified recovered mass with real generator–recycler pairs.

SCALE / OPERATE RELIABLY

Authentication, shared durable quotas, monitoring, dated source refreshes and regional data partnerships.

EXPAND / QUOTED ECONOMICS

Road-route distances, buyer-specific logistics quotes and auditable price comparisons. These are not live today.

Current ranking uses state/district proximity, not road kilometres or cheapest price. No verified matching or environmental benefit is implied.

Provide links to your:

FINAL PRODUCT LINK

https://wasterecast.web.app/

GITHUB PUBLIC REPOSITORY

https://github.com/anshuman542018/wasterecast

DEMO VIDEO LINK / 3 MINUTES

https://drive.google.com/file/d/1e1EyM-4qKs6WXAbR9nq35Z9UsCGghN5u/view

All links must be accessible without requesting access. Submit the PDF exported from this prescribed template.

Evidence, boundaries and evaluation readiness

Real records. Explicit uncertainty. A human-confirmed next step.

Source evidence	MPCB Directory 2025; TNPCB solvent list; WBPCB inventory 2023–24. Source pages and retrieval dates stay attached.
What a lead means	A facility worth checking. Listing status does not prove current authorization, demand, capacity or material acceptance.
What AI confidence means	An extraction estimate from the model. It is not a measured accuracy score or approval to move waste.
Evaluation access	Public prototype and repository. Participant confirmed public AI profiling; free-tier quota and provider availability still apply.

Google free-tier processing/logging terms apply. Avoid confidential uploads. No completed transactions or certified compliance are claimed.

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Thank you

